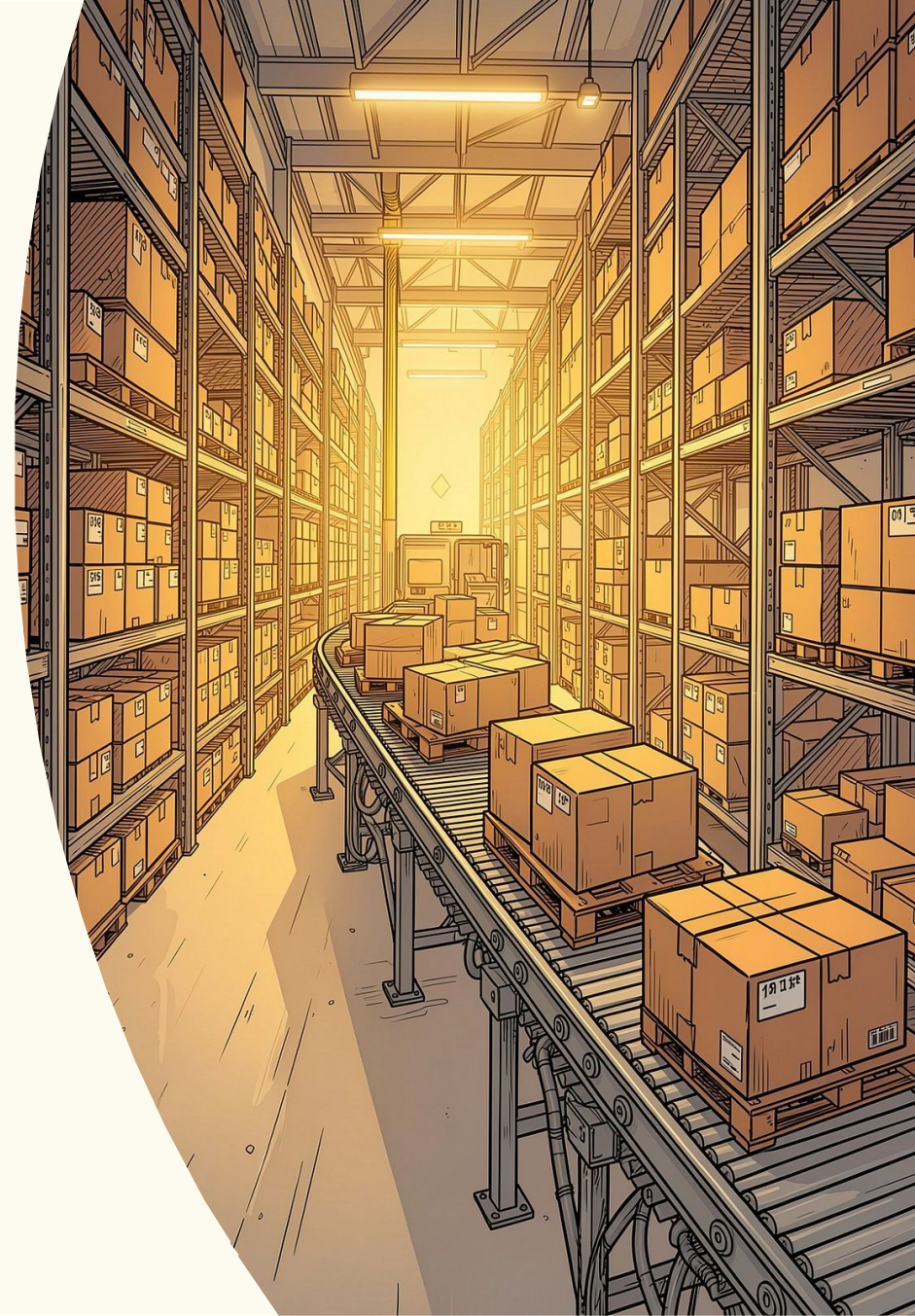


Improving E-Commerce Revenue & Customer Experience

The objective of this analysis is to identify sales opportunities and operational issues affecting revenue and customer satisfaction.

Brazilian E-Commerce Dataset (Olist) | 2016–2018



Data & Methodology

99K+

Orders analysed

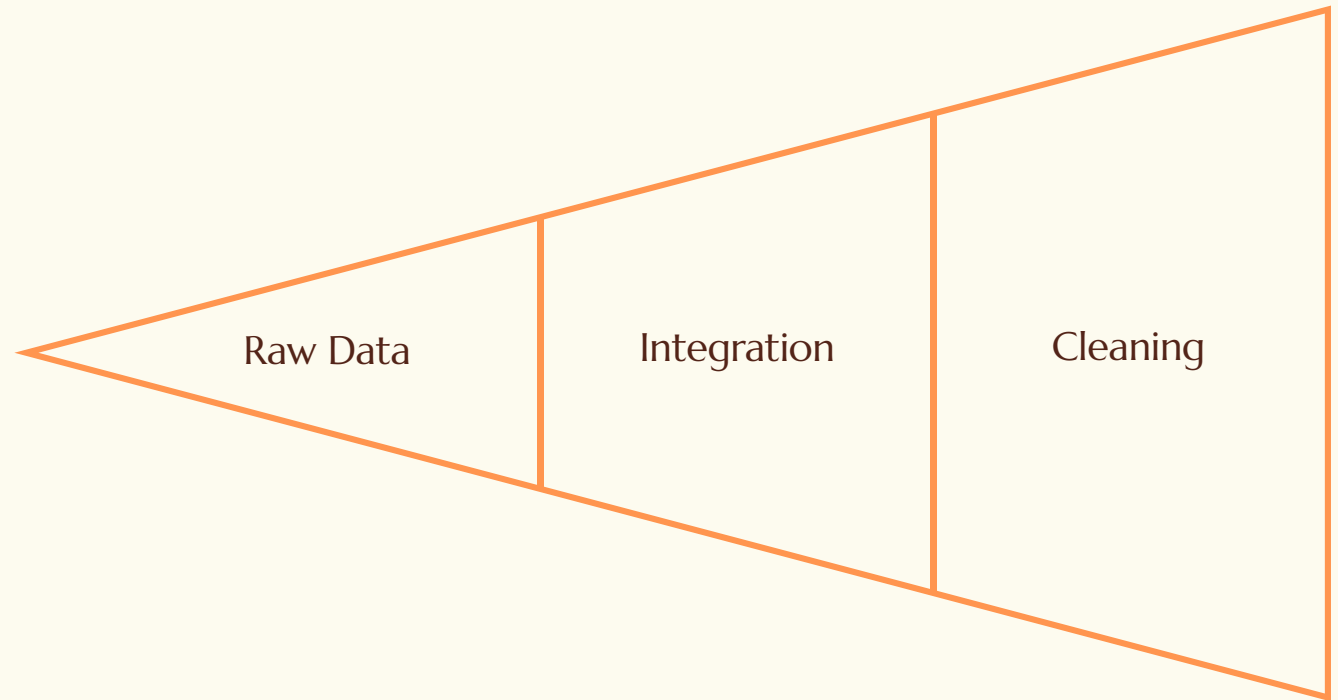
8+

Related tables integrated

2016–2018

Data time range

Analytical Pipeline



Key Findings

1

Delivery & Satisfaction

Delivery performance strongly correlates with customer satisfaction — on-time orders score significantly higher than late or undelivered ones.

2

Category Concentration

Revenue is concentrated in specific product categories, presenting clear prioritisation opportunities for commercial strategy.

3

Geographic Concentration

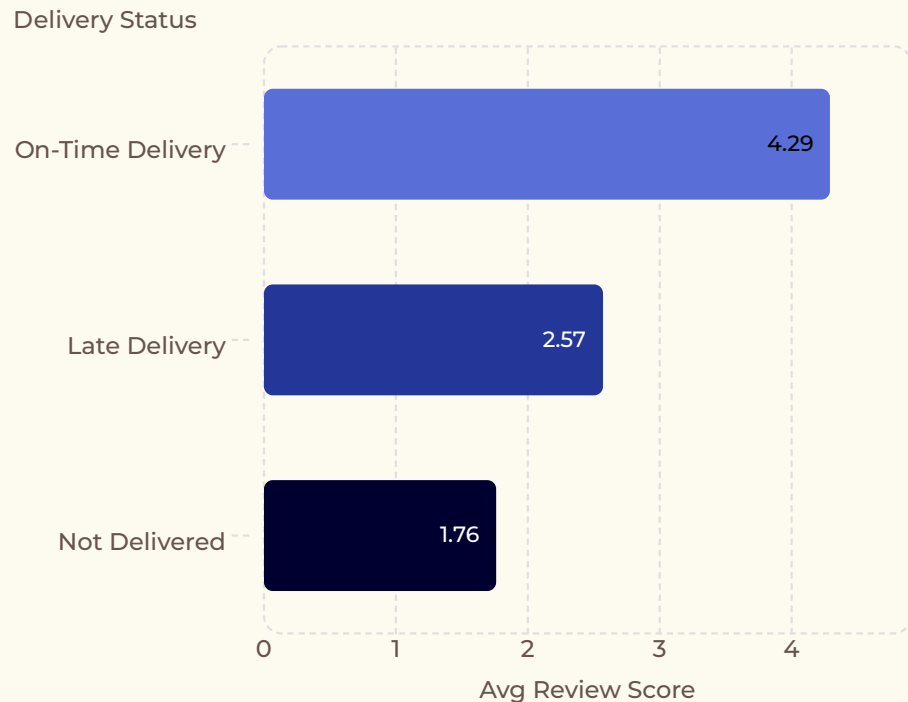
Sales and customer activity are concentrated geographically, with high-potential regions underserved relative to demand signals.

4

Seasonal Patterns

Sales show identifiable seasonal patterns that can inform inventory planning, marketing spend and promotional timing.

Deep Dive: Delivery Performance → Customer Satisfaction



★ Average customer review score by delivery status

On-Time Delivery

~**4.29** ★ — Customers who received orders on time rated their experience nearly 5 stars on average.

Late Delivery

~**2.57** ★ — A delay alone reduces satisfaction by more than 1.7 points, a significant drop.

Not Delivered

~**1.76** ★ — Non-delivery produces near-minimum scores, severely damaging brand trust.

Improving delivery reliability may have significant customer-experience benefits. ([GitHub](#))

Recommendations

1 Improve Delivery Monitoring

Invest in real-time logistics tracking, proactive delay alerts and carrier performance reviews to reduce late and failed deliveries.

2 Prioritise High-Performing Product Categories

Allocate marketing budget and inventory toward categories driving the majority of revenue; deprioritise or review low-converting segments.

3 Expand into High-Potential Regions

Target underserved geographic areas that show demand signals but low penetration with localised campaigns and logistics partnerships.



Expected Business Impact

Action	KPI
 Improve delivery monitoring	↓ Late delivery rate
 Improve category strategy	↑ Revenue
 Regional expansion	↑ Orders
 Improve customer experience	↑ Review score

These actions address the root causes identified in the analysis and are directly measurable through existing platform KPIs.

Limitations & Next Steps

Limitations

Historical Data

Dataset covers 2016–2018 only; market conditions may have shifted significantly since then.

Single Marketplace

Analysis is limited to the Olist platform; findings may not generalise across all e-commerce channels.

Anonymised Customers

Customer identity is masked, preventing cohort-level or CLV analysis.

Association $\neq$ Causation

Observed correlations (e.g. delivery $\rightarrow$ satisfaction) do not confirm causal direction without further testing.

Next Steps



Incorporate Newer Data

Refresh analysis with post-2018 transactional data to reflect current market behaviour.



Add Marketing Spend

Layer in campaign cost data to calculate channel ROI and attribute revenue more accurately.



Logistics-Provider Data

Integrate carrier-level performance records to pinpoint delivery failure root causes.



Customer Acquisition Costs

Include CAC to evaluate regional expansion viability and category profitability holistically.